

# An LLM-based knowledge and function-augmented approach for optimal design of remanufacturing process

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## ABSTRACT

Remanufacturing that returns used products to a like-new condition, is essential for promoting the circular economy and reducing carbon emissions. The optimal design of remanufacturing process (ODRP), as a knowledge-intensive complex decision-making task, plays a vital role in the success of remanufacturing. However, insufficient utilization of remanufacturing knowledge, and the trade-offs among multi-objectives in decision-making scenarios make ODRP time-consuming and labor-intensive. With the development of next-generation artificial intelligence (AI) technologies, large language models (LLMs) provide an important enabling tool for complex decision-making tasks. However, existing LLMs still face significant challenges in ODRP due to a lack of remanufacturing knowledge and computational capabilities. To address this issue, an LLM-based approach augmented with knowledge and function is proposed in this paper. Firstly, based on the establishment of remanufacturing process database, a retrieval augmented generation (RAG)-based knowledge-augmented strategy is designed to retrieve failure information (e.g., failure form, failure degree, etc.) of returned products through the interaction with LLMs, and generate the feasible remanufacturing schemes. Secondly, a function-augmented mechanism with function learning is also proposed to calculate each objective value and combined assessed value of the generated remanufacturing schemes with LLMs, assisting process designers in designing optimal remanufacturing scheme and process parameters. Finally, the proposed approach is validated using a case study on automobile gearbox remanufacturing. The results indicate that the proposed knowledge-augmented strategy improves the average accuracy from 65% to 79% when using ChatGLM3-6B as the base LLMs. Additionally, the proposed function-augmented mechanism can calculate the minimum combined assessed value and make more realistic results for ODRP. Meanwhile, the proposed integrated approach provides a solution to knowledge-intensive and complex decision-making tasks, which has a broad application prospect.

## 1. Introduction

As a strategic emerging industry in China, remanufacturing has become an important focus for promoting the construction of a circular economy and reducing carbon emissions [1]. For example, remanufacturing automotive components leads to approximately 88 % material savings, a 53 % reduction in CO<sub>2</sub> emissions, and 56 % less energy consumption compared to manufacturing new ones [2]. Considering these substantial environmental and economic benefits, the remanufacturing

industry is progressively implementing intelligent strategies to enhance efficiency, lower costs, and promote sustainability. ODRP, which bridges returned end-of-life (EOL) products and remanufactured products, plays a vital role in the success of remanufacturing [3]. However, due to the insufficient utilization of remanufacturing process knowledge, and the trade-offs involved in multi-objective decision-making scenarios, ODRP heavily depends on the skills and experience of process designers. This makes ODRP a time-consuming, labor-intensive, and subjective-influenced process [4], posing a major obstacle to the intelligent

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development of remanufacturing [5,6].

ODRP can be defined as a knowledge-intensive decision-making task with multi-stages and multi-objectives [7,8]. It not only needs to design feasible remanufacturing schemes based on the failure states of EOL products/components, but also requires to evaluate the objectives (e.g., time, cost, energy consumption, etc.) of these schemes, and design optimal scheme and process parameters [9]. The reuse of remanufacturing process knowledge is key to achieving ODRP, however, the remanufacturing process knowledge involves multiple disciplines, such as mechanical engineering, materials science, optimal design, and others. The knowledge base in these fields is extensive, and the process designers are required to have interdisciplinary and comprehensive capabilities. Furthermore, ODRP entails numerous decision-making stages with unclear interrelations. Conflicts and trade-offs exist between decision objectives, and changes in one stage or objective can trigger a chain reaction, potentially affecting the entire process. Most existing studies design remanufacturing process according to the failure characteristics of returned products [10,11]. Some others focus on selecting remanufacturing strategies and designing the process using evolutionary algorithms after the strategies are established [12,13]. These studies have undoubtedly laid a valuable foundation for ODRP. However, they are limited by the constraints of human knowledge and an incomplete understanding of the relationships between decision-making elements, which hinders the intelligent development of ODRP.

The advancement of the next-generation artificial intelligence (AI) technologies offers robust support in addressing the aforementioned challenges. In this context, Knowledge Graph (KG) has been demonstrated to effectively support the analysis, organization, and reasoning of manufacturing knowledge [14], which in turn reduces the reliance on manual operations during manufacturing process design [15,16]. However, KG still faces significant limitations, such as its insufficient capacity for multi-hop queries and reasoning tasks. These limitations hinder its ability to handle the complexity inherent in ODRP problems (e.g., process selection, route design, parameter setting, etc.) [17]. The development of LLMs provides new ideas for solving the challenges of ODRP. Their robust semantic comprehension and knowledge inference [18] can help address KG's shortcomings in handling complex decision-making within ODRP [19]. However, existing pre-trained LLMs lack specialized expertise in the remanufacturing domain [20] and the tool-calling ability to calculate decision values of different remanufacturing schemes [21]. These limitations may lead to inaccurate decision-making results, making it difficult to provide the optimal choice of remanufacturing schemes and process parameters for process designers. The emergence of RAG and function learning addresses the inherent limitations of existing LLMs. Specifically, interacting with RAG and function learning can enhance LLMs' remanufacturing process knowledge and standard computational capabilities, thereby further improving LLMs' ability to handle multi-objective decision-making within ODRP.

With the purpose of efficiently reusing remanufacturing process knowledge and fostering the intelligent development of ODRP. This paper proposes an LLM-based approach augmented with knowledge and function for ODRP, including the establishment of a remanufacturing process database, a RAG-based knowledge-augmented strategy, and a function-augmented mechanism with function learning. The main contributions of this study are as follows:

- (1) An LLM-based approach augmented with knowledge and function is proposed for ODRP, solving the problem of lack of remanufacturing domain knowledge and standardized computational capabilities of LLMs. This approach can be also extended to knowledge-intensive decision-making tasks to support the implementation of LLMs in industry.
- (2) A RAG-based knowledge-augmented strategy is designed to retrieve relevant information and generate feasible remanufacturing schemes through interacting with the remanufacturing process database and LLMs.

- (3) A function-augmented mechanism with function learning is employed to help LLMs to call the relevant decision objective formulas, e.g., energy consumption, cost, and calculate the combined assessed values of each feasible remanufacturing scheme, which is helpful to assist process designers in designing optimal remanufacturing scheme and process parameters.

The rest of the paper is organized as follows: Section 2 provides related works; Section 3 describes the proposed approach and its general framework. Section 4 presents a case study to validate the effectiveness of the proposed approach. Finally, Section 5 summarizes the paper and discusses potential directions for future research.

## 2. Related works

### 2.1. Optimal design of remanufacturing process

In common, ODRP is started with the failure state of the returned products or components, and determine the optimal remanufacturing scheme and process parameters considering some decision objectives, such as cost reduction, energy savings, and efficiency improvement, etc. Many scholars have conducted research on this issue and achieved significant results. Lahmar et al. [22] developed a multi-objective mathematical model focusing on economic and environmental factors, utilizing the NSGA-II algorithm to determine optimal remanufacturing plans. Zhang et al. [23] developed a multi-objective optimization design model for remanufacturing process selection, considering carbon emissions, time, and cost, and solved this model by using the differential evolution and grey wolf optimization algorithm. Stamer et al. [24] modeled the remanufacturing planning process as a Markov process, which enables the evaluation of all Pareto optimal process plans in terms of cost efficiency and reliability. The above work highlights that the ODRP is a complex decision-making problem, requiring consideration of remanufacturing process aspects and parameters, but most of existing studies have concentrated on specific or limited scenarios. It does not fully cover the complexity remanufacturing processes, thereby preventing the achievement of a global optimized solution of ODRP.

ODRP is also considered a highly knowledge-intensive task, requiring the acquisition of knowledge related to design specifications, failure characteristics, and the capabilities of remanufacturing facilities, etc. Over time, remanufacturing enterprises have accumulated a large number of process cases, which provide a valuable resource for process design. However, the knowledge required for the remanufacturing process is dispersed across various case databases, process profiles, maintenance records, and other documents from different enterprises, effectively organizing and reusing this knowledge remains a significant challenge. As a result, several scholars have conducted in-depth studies on this topic. He et al. [25] proposed an ontology-based unified modeling approach for remanufacturing process knowledge and applied instance reasoning techniques to ODRP. Ke et al. [26] proposed a performance requirement-based method for integrating remanufacturing processes, which reuses historical data to generate process schemes. Egbe et al. [27] proposed a framework based on case-based reasoning to develop CAPP systems for remanufacturing process planning, treating implementation experience as a knowledge resource for decision-making. Although these studies alleviate the burden on process personnel and enhance decision-making efficiency through knowledge reuse, they still have limitations when addressing complex ODRP problems that involve multiple stages and objectives.

### 2.2. LLMs for decision-making

LLMs possess exceptional adaptability to various scenarios, the ability to integrate knowledge across domains, and efficient human-machine interaction capabilities. These strengths enable them to overcome decision-making limitations in ODRP, reduce the reliance on

manual operations, and promote the automation and intelligence of process design. According to the characteristics of ODRP, studies have been conducted on knowledge-intensive complex decision-making tasks using LLMs.

Some scholars have proposed knowledge-augmented strategies to address the lack of specialized domain knowledge in LLMs, mainly including RAG and fine-tuning. Among them, RAG is used to improve the quality and accuracy of generated content by combining the advantages of retrieval systems with LLMs. Aftab et al. [28] focused on enhancing prompts using RAG to guide LLMs in generating meaningful responses for biomedical QA tasks. Liang et al. [29] inspired by the Knowledge-Based Visual Question Answering (KB-VQA) domain, improve the answering capabilities of existing MedVQA models for medical visual questions using RAG. Zhu et al. [30] proposed an effective query system for coal mine safety information security and developed a rapid, engineering-based coal mine safety Q&A dataset to guide RAG implementation in this domain. Liu et al. [31] focused on enhancing RAG for processing complex automotive industry documents using locally deployed Ollama models, offering a practical guide for deploying a local RAG system. Fine-tuning is a strategy to adjust model parameters to enhance the specialized capabilities of LLMs. Guthrie et al. [32] aimed to fine-tune a LLM for domain-specific text generation in surgical and anesthesia residency education. Wu et al. [33] proposed a pre-trained LLMs based on biomedical literature, which was fine-tuned by incorporating medical knowledge into the LLaMA model, thereby improving its performance in a medical Q&A benchmark test. Zhu et al. [34] introduced an innovative approach for predicting the content of aflatoxin B1 (AFB1) by fine-tuning the Babbage-002 LLM. Latif et al. [35] fine-tuned GPT-3.5 for automatic scoring of student responses on domain-specific data in education. It should be noted that fine-tuning is costly in terms of computational resources and energy consumption, and it requires high-performance hardware [36]. The emergence of RAG reduces the computational burden of traditional fine-tuning strategies and decreases reliance on high-end hardware, and is more suitable for application and implementation in remanufacturing enterprises.

Some scholars have proposed function-augmented mechanisms for LLMs to handle complex decision-making tasks [37]. Li et al. [38] proposed a novel agent framework for LLMs, equipped with specialized tools to improve strategic decision-making. This framework suggests the potential future development of LLMs applications in interactive environments. Lim et al. [39] proposed a framework for a multi-agent manufacturing system integrated with LLMs, which enhances the adaptability and coordination of LLMs in manufacturing by improving their natural language processing capabilities. Khoei et al. [40] automated complex data analysis tasks with decision support by integrating LLMs agents with external tools, thus significantly improving the efficiency and accuracy of software release decisions.

To sum up, ODRP has received a wide range of attention, but existing research is often limited to specific or limited scenarios, lacking a comprehensive understanding of the interconnections between decision-making objectives and remanufacturing processes. LLMs are widely concerned to offer a new perspective for solving knowledge-intensive complex decision-making problems. In particular, the emergence of new technologies such as RAG and function learning provides solutions to enhance the specialized domain knowledge and standardized computational capabilities of LLMs. Therefore, the focus of this study is on how to leverage these new technologies to overcome the inherent shortcomings of LLMs and assist process designers in performing the ODRP.

### 3. The overall research framework

An LLM-based approach augmented with knowledge and function is developed to efficiently reuse remanufacturing process knowledge and handle multi-objective decision-making within ODRP. This approach can help process designers design optimal remanufacturing scheme and

process parameters, reducing their workload. Additionally, it enhances remanufacturing efficiency and quality, promoting the intelligent development of ODRP. The overall research framework includes three stages: (1) establishing a remanufacturing process database, (2) developing a knowledge-augmented strategy for generating feasible remanufacturing schemes, and (3) designing a function-augmented mechanism for optimal scheme and parameters decision-making, as illustrated in Fig. 1.

- (1) Establishing a remanufacturing process database. In stage 1, a remanufacturing process database is established, which includes two components: a knowledge database and a function database. The knowledge database is constructed to enhance the remanufacturing domain knowledge of LLMs to generate feasible remanufacturing schemes in stage 2. The function database is constructed to provide computational tools for ODRP decision-making for LLMs in stage 3.
- (2) Developing a knowledge-augmented strategy for generating feasible remanufacturing schemes. In stage 2, a RAG technology is employed to retrieve relevant information to generate feasible remanufacturing schemes through the interaction between LLMs and the knowledge database established in stage 1.
- (3) Designing a function-augmented mechanism for optimal scheme and parameters decision-making. In stage 3, a function learning technology is designed to calculate the combined assessed values of each feasible remanufacturing scheme by using the interaction between LLMs and the function database established in stage 1, assisting process designers in designing the optimal remanufacturing scheme and process parameters.

#### 3.1. Establishing a remanufacturing process database

##### 3.1.1. Remanufacturing process knowledge database

The remanufacturing process knowledge database is established to enhance specialized remanufacturing domain knowledge for LLMs by using RAG. In remanufacturing enterprises, the remanufacturing process typically involves multiple stages such as product design, disassembly, cleaning, inspection, repair, assembly, and testing, and the related remanufacturing process knowledge may be embedded in different documents with diverse types and formats, as shown in Table 1.

In Table 1, the remanufacturing process knowledge is embedded in these multi-source and heterogeneous documents. It is important to note that the most important information of ODRP, such as remanufacturing equipment and tools, remanufacturing schemes, remanufacturing process parameters, etc., is mostly represented as textual data in these files. Compared to other external data sources, the paradigm of integrated textual data can cover a wide range of domains, enabling LLMs to address interdisciplinary problems [41,42]. Additionally, this textual data can provide richer contextual information, ensuring that LLMs generate more accurate responses. Therefore, this study mainly focuses on extracting and segmenting the textual data from the remanufacturing process documents to establish a vectorized embedding model for the establishment of knowledge database, so that LLMs can interact with the knowledge database to generate a set of feasible remanufacturing schemes by RAG.

##### 3.1.2. Remanufacturing process function database

The remanufacturing process function database is constructed to provide a calculation basis for the values of decision objectives, such as energy consumption, cost, etc., which is the prerequisite for optimal scheme and parameters decision-making with function learning. It is important to note that the remanufacturing schemes consist of different work steps, and each work step contains specific parameters and calculation formulas for decision objective values. Therefore, this paper compiles the methods for common decision objectives associated with

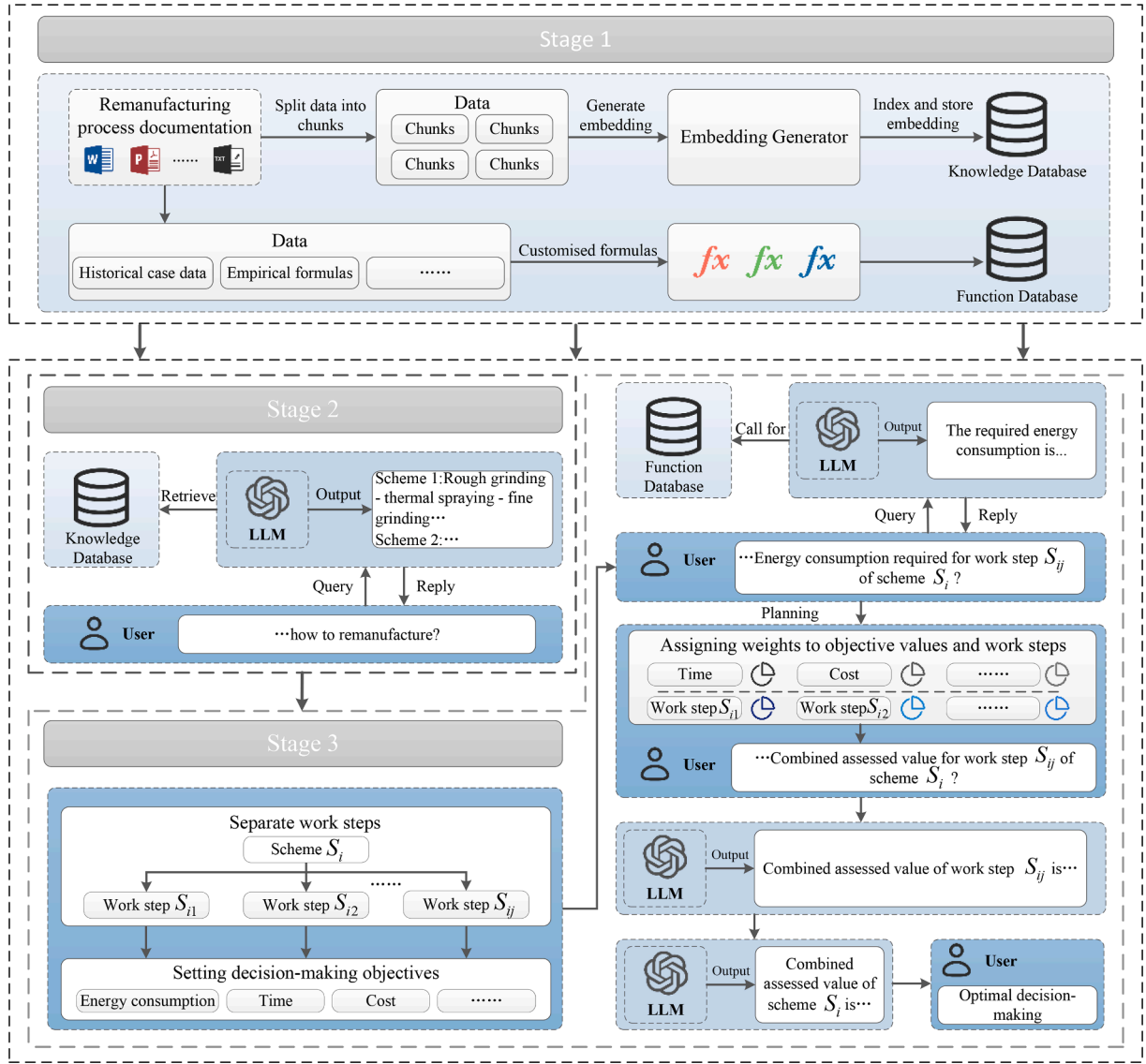


Fig. 1. Overall framework for an LLM-based approach augmented with knowledge and function for ODRP.

typical remanufacturing work steps and establishes a function database for these calculations, as described in Table 2.

In Table 2,  $X_r = (x_r^1, x_r^2, \dots, x_r^{m_r})$  represents the remanufacturing parameter set containing  $m_r$  parameters of a specific work step. And  $Y_r = (y_r^1, y_r^2, \dots, y_r^{n_r})$  is the decision objective set containing  $n_r$  objectives of this specific work step, which can be expressed as functions of the parameters, as shown in Eq. (1).

$$y_r^{n_r} = f_{n_r}(x_r^1, x_r^2, \dots, x_r^{m_r}) \quad (1)$$

where  $y_r^{n_r}$  denotes the  $n_r$ th decision objective in this specific work step, and  $f_{n_r}(x_r^1, x_r^2, \dots, x_r^{m_r})$  is the corresponding formula. As an example, the function database calculates the cutting energy consumption function, and the pseudo-code for generating the objective value is shown in Fig. 2.

### 3.2. Developing a knowledge-augmented strategy for generating feasible remanufacturing schemes

RAG combines the advantages of retrieval system and LLMs, enabling the effective reuse of accumulated case knowledge and a better understanding of user semantics. When users pose a question, the

retrieval system vectorizes the question and searches for relevant knowledge texts in the knowledge database. Finally, RAG combines the retrieved information with the user's question and uses LLMs to generate content with enhanced quality and accuracy. Therefore, a RAG-based knowledge-augmented strategy is proposed to generate feasible remanufacturing schemes. Specifically, based on the remanufacturing process knowledge database constructed in Section 3.1.1, the implementation of RAG in this paper includes two parts: remanufacturing process knowledge retrieval system and prompt template-augmented generation. The process for generating the remanufacturing schemes is shown in Fig. 3.

First, a knowledge database space  $o$  is defined, which contains the set  $o = \{O_1, O_2, \dots, O_n\}$  of text blocks of documents that have been embedded and the set  $o^* = \{O_1^*, O_2^*, \dots, O_n^*\}$  of their corresponding text block vectorization results, and then the process designers' question  $q$  is embedded and vectorized to be represented as  $O_q$ .

Next, a similarity measure  $S$  is chosen to calculate the similarity between the question vector  $O_q$  and the text block vector  $O_i^*$ . The similarity measure method takes the cosine similarity  $\cosine(O_q, O_i^*)$ , as shown in Eq. (2).

$$\cosine(O_q, O_i^*) = \frac{O_q \cdot O_i^*}{\|O_q\| \|O_i^*\|} \quad (2)$$



**Table 1**

Documentation of remanufacturing processes knowledge sources.

| Type                          | Format                    | Description                                                                                                                                   |
|-------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Technical standards (TS)      | PDF, Word                 | Remanufactured processes and products should adhere to, such as ISO standards and industry-specific standards.                                |
| Process flow diagrams (PFD)   | Visio, PDF, JPEG, PNG     | The process flow from EOL products to remanufactured products.                                                                                |
| Work instructions (WI)        | Word, PDF                 | The specific operational steps for each process, such as remanufacturing equipment and tools chosen, remanufacturing parameters setting, etc. |
| Bill of materials             | Excel, CSV                | Lists all raw materials, parts, and their quantities used in remanufacturing processes.                                                       |
| Equipment maintenance manuals | PDF, Word                 | Provides maintenance guidelines for all equipment used in the remanufacturing process.                                                        |
| Training materials            | PowerPoint, PDF, Video    | Training materials provided to employees regarding remanufacturing technologies and safety regulations.                                       |
| Project management documents  | Project, Excel, Word, PDF | Documents related to historical remanufacturing cases, such as project plan documents, progress reports, risk management documents, etc.      |

**Table 2**

The description of the remanufacturing process function database.

| Remanufacturing stages | Work steps         | Process parameters                                           | Objective formulas                                           |
|------------------------|--------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| Mechanical processing  | Cutting            | $X_1 = (x_1^1, x_1^2, \dots, x_1^{m_1})$                     | $Y_1 = (y_1^1, y_1^2, \dots, y_1^{n_1})$                     |
|                        | Grinding           | $X_2 = (x_2^1, x_2^2, \dots, x_2^{m_2})$                     | $Y_2 = (y_2^1, y_2^2, \dots, y_2^{n_2})$                     |
| Surface repair         | .....              | .....                                                        | .....                                                        |
|                        | Laser cladding     | $X_k = (x_k^1, x_k^2, \dots, x_k^{m_k})$                     | $Y_k = (y_k^1, y_k^2, \dots, y_k^{n_k})$                     |
|                        | Spray coating      | $X_{k+1} = (x_{k+1}^1, x_{k+1}^2, \dots, x_{k+1}^{m_{k+1}})$ | $Y_{k+1} = (y_{k+1}^1, y_{k+1}^2, \dots, y_{k+1}^{n_{k+1}})$ |
| Disassembly            | .....              | .....                                                        | .....                                                        |
|                        | Manual disassembly | $X_{r-1} = (x_{r-1}^1, x_{r-1}^2, \dots, x_{r-1}^{m_{r-1}})$ | $Y_{r-1} = (y_{r-1}^1, y_{r-1}^2, \dots, y_{r-1}^{n_{r-1}})$ |
|                        | .....              | $X_r = (x_r^1, x_r^2, \dots, x_r^{m_r})$                     | $Y_r = (y_r^1, y_r^2, \dots, y_r^{n_r})$                     |

To perform the computation of similarity scores, the similarity score between each text block vector  $O_i^*$  and the question vector  $O_q$  is computed as  $s_i$ :  $s_i = S(O_q, O_i^*)$ . The closer the computed value is to 1 means that the higher the similarity between  $O_i^*$  and  $O_q$ , the higher the

corresponding similarity score of  $O_i^*$  will be.

Then, a positive integer  $k$  is chosen to retrieve the  $k$  text block vectors that are most relevant to the  $O_q$ . This can be achieved by sorting the similarity scores  $s_i$  and selecting the  $k$  matching vectors with the higher scores.

Finally, based on the  $k$  matching vectors  $O_{q_i}^*$ , retrieving  $k$  corresponding text block element information  $O_{q_i}$  from the  $o$  to form a prompt template with the process designers' question  $q$  to enhance the LLMs in generating feasible remanufacturing schemes.

Compared to relying solely on the data within LLMs, providing external information on remanufacturing processes through proposed knowledge-augmented strategy can significantly enhance the quality of remanufacturing schemes generated by LLMs. This strategy results in more detailed and objective responses that better align with specific scenario requirements.

### 3.3. Designing a function-augmented mechanism for optimal scheme and parameters decision-making

The function-augmented mechanism with function learning is proposed for optimal scheme and parameters decision-making. Its function learning technology consists of three phases: function selection, function calling, and optimal decision-making. All phases are in the form of prompt templates for LLMs to learn relevant competencies. The process of augmenting LLMs with function learning for ODRP is shown in Fig. 4. Regarding the selection of LLMs, ChatGLM3-6B offers significant advantages in terms of scale, performance, adaptability, and flexibility compared to other open-source models, making it an ideal foundational model. Therefore, this section uses ChatGLM3-6B as the foundational model for all LLM-based functional demonstrations.

#### 3.3.1. Function selection

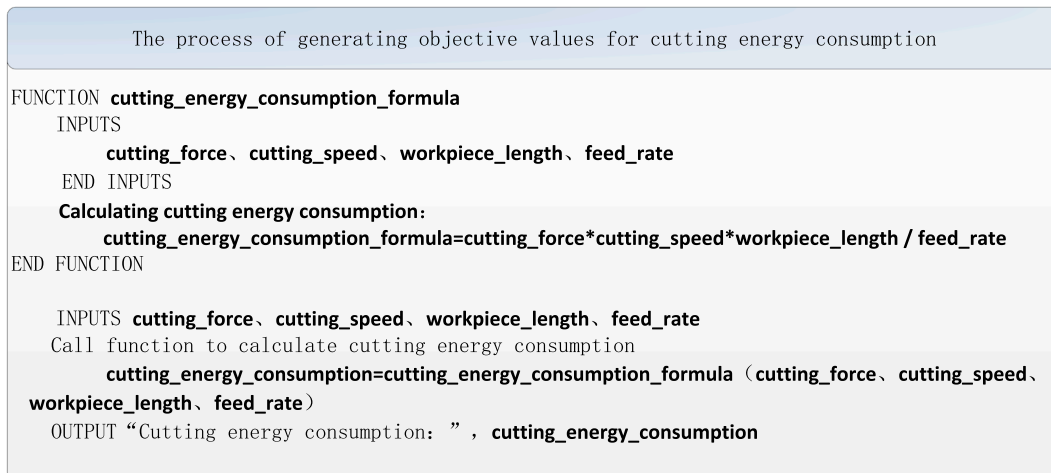
After obtaining the feasible remanufacturing schemes, the process designers divide the work steps for each remanufacturing scheme and set several decision objectives in conjunction with the decision requirements, as follows:

In general, the feasible remanufacturing schemes are divided into several schemes, and each scheme consists of several work steps. The set of work steps for each remanufacturing scheme  $S_i$  is further obtained as shown in Eq. (3).

$$S_i = \{S_{i1}, S_{i2}, \dots, S_{ij}\} \quad (3)$$

where  $S_{ij}$  denotes the  $j$ th work step in the remanufacturing scheme  $S_i$ .

Then, the process designers uniformly set the decision objectives for each scheme based on the remanufacturing requirements, and the set of

**Fig. 2.** The process of generating objective values for cutting energy consumption.

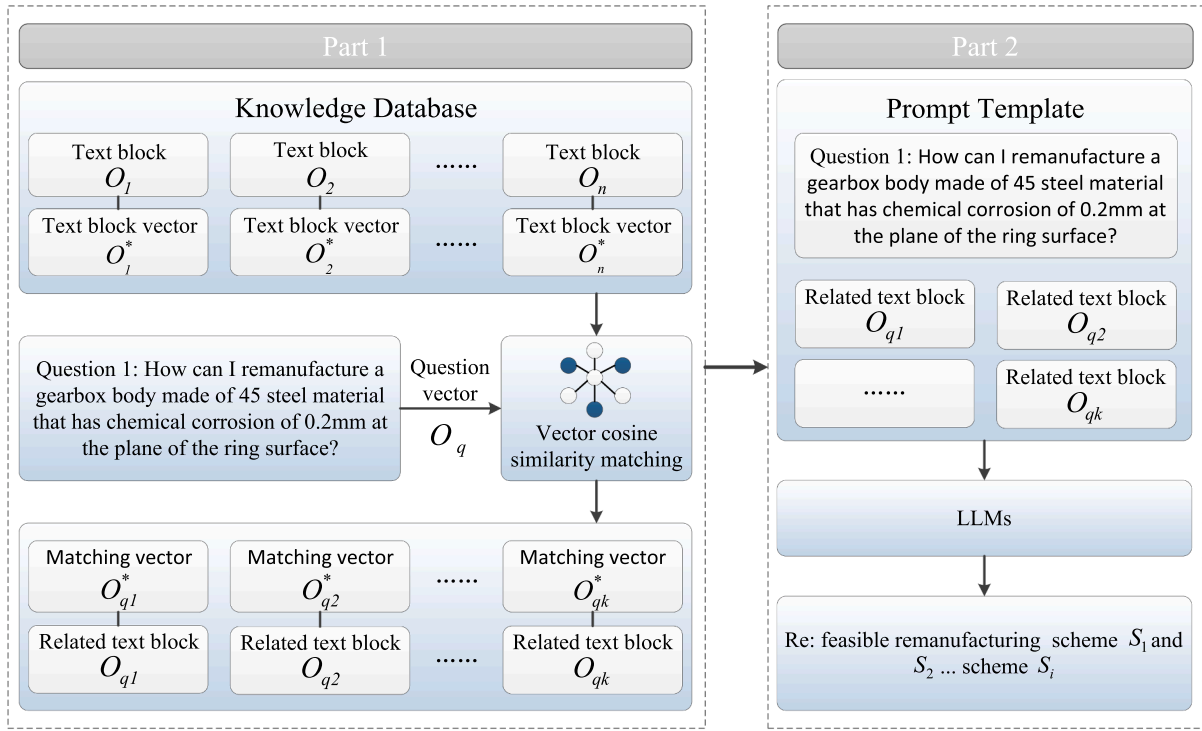


Fig. 3. The process for generating feasible remanufacturing schemes.

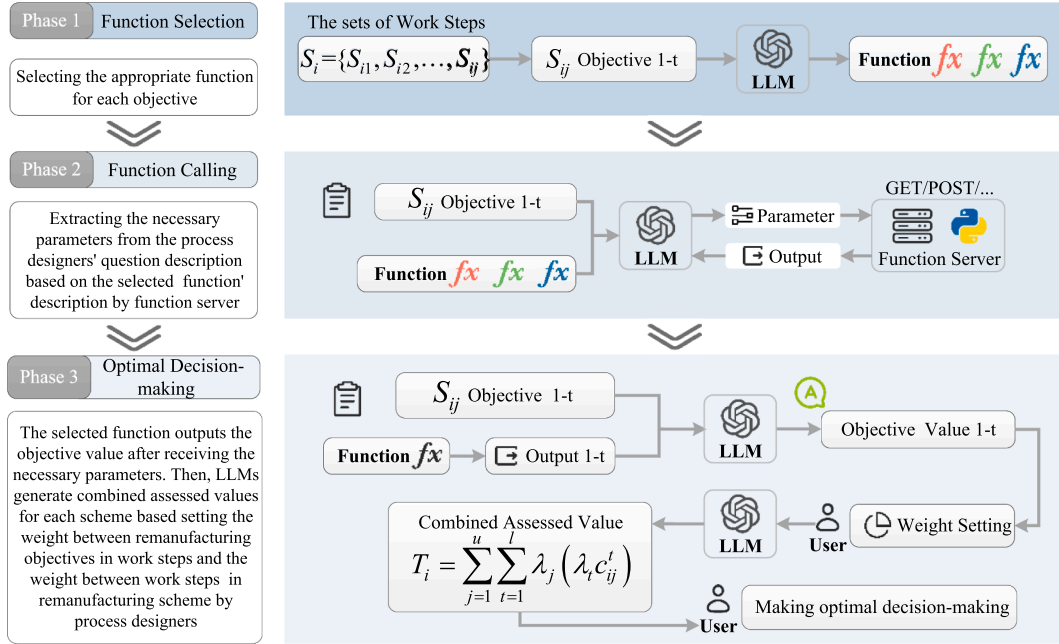


Fig. 4. The process of augmenting LLMs with function learning for ODRP.

decision objectives for the work step  $S_{ij}$  can be obtained as shown in Eq. (4).

$$B_{ij} = (b_{ij}^1, b_{ij}^2, \dots, b_{ij}^t) \quad (4)$$

where  $b_{ij}^t$  is the  $t$ th decision objective of the work step  $S_{ij}$ .

After the process designers set the decision objectives, multiple decision objective values for any work step  $S_{ij}$  of the scheme  $S_i$  need to be calculated, and the key to this process is to select the appropriate function. In this phase, a function selection method based on LLMs is

adopted, which requires LLMs to select the most appropriate function based on the description of the function and the process designers' specific question. Taking the calculation of grinding energy consumption as an example, Table 3 shows the process of performing function selection using the ChatGLM3-6B as the base LLMs.

In the example, we can observe that ChatGLM3-6B can accurately identify and select appropriate functions for the process designers' question of grinding energy consumption. Specifically, when faced with a large number of potential candidate functions, ChatGLM3-6B can efficiently filter out the specific function that is most suitable for

**Table 3**

An Example of Function Selection with ChatGLM3-6B.

| An Example of Function Selection with ChatGLM3-6B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instruction Prompt:</b> You are currently in the function selection phase. You are given candidate functions that can potentially solve the user's question. Among candidate functions, select a relevant function that would help solve the user's question.<br><b>Objective:</b> The machining length is known to be 1.0 mm and the spindle speed is 1500.0 RPN. The grinding speed for rough grinding is 60.0 m/min, the feed rate is 150.0 mm/rad, and the grinding depth is 1.0 mm. Calculate the grinding energy consumption required for this.<br><b>Candidate Functions:</b> 1. Grinding energy consumption formula: Calculate the energy consumption of the grinding operation based on the grinding speed, feed rate, grinding depth, workpiece length, and spindle speed. 2. Cutting energy consumption formula: Calculate the energy consumption for a cutting operation based on cutting force, cutting speed, workpiece length, feed rate. 3. Spraying energy consumption formula: .....<br><b>Output:</b> Grinding energy consumption formula: Calculate the energy consumption of the grinding operation based on the grinding speed, feed rate, grinding depth, workpiece length, and spindle speed. |

answering the process designers' question by deeply analyzing the characteristics of each function.

### 3.3.2. Function calling

During the execution of the function calling phase, the task of LLMs is to accurately extract the necessary parameters from the process designers' question description, based on the parameter specifications detailed in the function description, and use them to initiate a data request to the function server. To ensure the accuracy and efficiency of this process, the LLMs must identify and extract the exact content of the parameters and their formatting, and must strictly follow the established output specifications to avoid generating any non-essential text. Table 4 shows an exemplary case of grinding energy function calling using the ChatGLM3-6B.

In this example, we can observe that the ChatGLM3-6B can accurately extract the key parameters required to execute the function calling based on the process designers' question and the description of the function related to grinding energy consumption. Specifically, the ChatGLM3-6B can extract key information from the function documentation and accurately identify the required parameter set to ensure the correct and effective execution of the function calling.

### 3.3.3. Optimal decision-making

Given the diversity and complexity of function outputs, presenting these results directly to the process designers is often inappropriate. Upon receiving the output data from a function, LLMs need to combine the process designers' question and incorporate their knowledge to construct a comprehensive and accurate response. For example, the ChatGLM3-6B follows this process when generating a response, ensuring that the output is accurate and easily understood by the process designers. Table 5 shows how the ChatGLM3-6B generates responses:

In this example, we can observe that the ChatGLM3-6B can accurately generate the objective value of grinding energy consumption after a series of operations such as function selection and function calling.

Afterward, assuming that multiple objective values for the work steps  $S_{ij}$  in each scheme  $S_i$  are computed based on function learning technology, the set of objective values  $C_{ij}$  for the work step  $S_{ij}$  can be obtained shown in Eq. (5).

$$C_{ij} = (c_{ij}^1, c_{ij}^2, \dots, c_{ij}^t) \quad (5)$$

where  $c_{ij}^t$  is the  $t$ th objective value in the work step  $S_{ij}$ .

In ODRP, the process designers should consider not only the trade-offs of different objectives in specific work step, but also need to consider the trade-offs of different work steps in the same scheme. In this paper, two levels of weights are set for calculating the combined assessed value of remanufacturing scheme based on the judgement of the process designer, as shown in Eq. (6).

$$T_i = \sum_{j=1}^u \sum_{t=1}^l \lambda_j (\lambda_t c_{ij}^t) \quad (6)$$

where  $T_i$  is the combined assessed value of remanufacturing scheme  $S_i$ ,  $\lambda_t$  and  $\lambda_j$  denote the weight between remanufacturing objectives in work steps  $S_{ij}$ , and the weight between work steps  $S_{ij}$  in the remanufacturing scheme  $S_i$ , respectively.

Finally, the process designers can use the  $T_i$  to design the optimal remanufacturing scheme and process parameters for ODRP.

## 4. Case study

As a core component of transmission system, the gearbox is widely used in automotive products, especially in mid-sized passenger vehicles with considerable ownership. The automotive gearbox will inevitably undergo wear, performance degradation and failure in long-term use, resulting in a decline in automotive performance. The remanufacturing of gearboxes will significantly reduce the demand for raw materials and energy, offering substantial economic and environmental benefits. However, due to the insufficient utilization of remanufacturing process knowledge for gearboxes and the trade-offs of multi-objectives in decision-making scenarios, the ODRP for gearboxes heavily relies on the skills and experience of process designers. Therefore, a case study was conducted on the ODRP of a specific type of automotive gearbox in this paper.

Through the application of the proposed approach, we first researched several enterprises' historical gearbox remanufacturing cases as well as various documents, including technical standards,

**Table 4**

An Example of Function Calling with ChatGLM3-6B.

| An Example of Function Calling with ChatGLM3-6B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instruction Prompt:</b> You are currently in the function calling phase. We have provided you with selected functions that may be used to solve the user's question. Your goal is to extract the required parameters to call the function from the user's question, based on the function description. The output is in the following format: {parameter name: parameter, ..., parameter name: parameter}.<br><b>Objective:</b> The machining length is known to be 1.0 mm and the spindle speed is 1500.0 RPN. The grinding speed for rough grinding is 60.0 m/min, the feed rate is 150.0 mm/rad, and the grinding depth is 1.0 mm. Calculate the grinding energy consumption required for this.<br><b>Selected Functions:</b> Function Name: {Grinding energy consumption formula}. Function description: {Calculate the energy consumption of the grinding operation based on the grinding speed, feed rate, grinding depth, workpiece length, and spindle speed.} Required params: { [name: processing lengths, type: FLOAT, description: Enter the value of processing length., name: spindle speed, type: FLOAT, description: Enter the value of spindle revolutions., name: grinding speed, type: FLOAT, description: Enter the value of grinding speed., name: feed rate, type: FLOAT, description: Enter the value s for feed speeds., name: grinding depth, type: FLOAT, description: Enter the value for grinding depth.] }<br><b>Output:</b> {processing lengths: "1.0", spindle speed: "1500.0", grinding speed: "60.0", feed rate: "150.0", grinding depth: "1.0"} |

**Table 5**

An Example of Response Generation with ChatGLM3-6B.

| An Example of Response Generation with ChatGLM3-6B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instruction Prompt:</b> You are currently in the response generation phase. You are given the output of a selected function that can be potentially used to solve the task. Your goal is to generate a response to the user's question based on the results from the function and your internal knowledge.<br><b>Objective:</b> The machining length is known to be 1.0 mm and the spindle speed is 1500.0 RPN. The grinding speed for rough grinding is 60.0 m/min, the feed rate is 150.0 mm/rad, and the grinding depth is 1.0 mm. Calculate the grinding energy consumption required for this.<br><b>Return from Function Calling:</b> The energy required for grinding is 13500000.<br><b>Output:</b> The required grinding energy consumption is calculated as 13500000. |

process flow diagrams, work instructions, etc., and constructed a remanufacturing process knowledge database of gearboxes by embedding model. Concurrently, we compiled formulas for calculating remanufacturing decision objectives including energy consumption, cost, time, etc., and constructed a function database. Building on this foundation, we designed a RAG-based knowledge-augmented strategy to retrieve failure information (e.g., failure form, failure degree, etc.) of the returned gearboxes through the interaction with ChatGLM3-6B, thereby generating feasible remanufacturing schemes. Additionally, we designed a function-augmented mechanism with function learning to calculate each objective value and combined assessed value of the generated remanufacturing schemes with ChatGLM3-6B, assisting process designers in designing optimal remanufacturing scheme and process parameters. The application of the approach proposed relies on the upper computer configured with an NVIDIA GeForce RTX 4070 graphics card, an i7-13700KF processor, and a 32 GB RAM, to ensure the stability and smoothness of system operation.

To evaluate the performance of the proposed approach in the aforementioned scenarios, we conducted an experiment from two perspectives: results and discussion. The results involved applying the proposed approach to a specific gearbox's ODRP to demonstrate its practical feasibility. The discussion validated the efficacy of each sub-method within the proposed approach. This included the evaluation of the knowledge-augmented strategy in reusing remanufacturing process knowledge and the evaluation of various phases of the function-learning mechanism, such as the accuracy of function selection, function calling, and optimal decision-making. The comprehensive evaluation above highlights the overall reliability and effectiveness of the approach.

#### 4.1. Results

We selected an automotive gearbox made of 45# steel with a chemical corrosion of 0.2 mm at the plane of the ring surface, which is an uncommon level of corrosion in automotive gearboxes, for a detailed

case study. The following are the results obtained from the knowledge-augmented strategy and the function-augmented mechanism, respectively.

##### 4.1.1. Results of the knowledge-augmented strategy for generating feasible remanufacturing schemes

The question "How to remanufacture an automobile gearbox made of 45# steel with chemical corrosion of 0.2 mm at the plane of the ring surface?" is used as input in Step 1 to generate two feasible remanufacturing schemes, with the results shown in Fig. 5.

The results indicate that, in the first step of process design, the feasible remanufacturing schemes generated using knowledge-augmented strategy closely matches the corresponding schemes found in the enterprise's historical remanufacturing data. This indicates that the proposed strategy effectively generates feasible remanufacturing schemes based on local textual data with detailed parameters.

##### 4.1.2. Results of the function-augmented mechanism for optimal scheme and parameters decision-making

Conduct the experiments of this section based on the generated schemes mentioned above. First, to simplify the analysis process for easier understanding and evaluation of the feasibility of the approach presented in this paper, assume that the weights  $\lambda_j$  between work steps  $S_{ij}$  in the remanufacturing scheme  $S_i$  are equally distributed. In this case, it is sufficient to consider the total value of each objective for a given scheme. Then, following the actual consideration of objective values in corporate remanufacturing, "the time, cost, and energy consumption of a certain scheme are known and the corresponding objective weight  $\lambda_t^1 : \lambda_t^2 : \lambda_t^3 = 4 : 4 : 2$ " is used as the question input in step 2 to calculate the combined assessed values of different schemes, and the results are shown in Table 6.

The results indicate that, in the second step of process design, the function-augmented mechanism can accurately calculate each objective

**Question:** How to remanufacture an automobile gearbox made of 45# steel with chemical corrosion of 0.2 mm at the plane of the ring surface?

**ChatGLM3-6B+ Knowledge Database**  
**Q&A:** Based on known information, we offer two remanufacturing options for a 0.2mm chemical corrosion problem at the plane of the ring surface of a gearbox body made of 45 steel:  
**Scheme 1:** Rough grinding, thermal spraying and fine grinding. The technologies involved are thermal spraying and grinding, and the equipment used are grinding machines and thermal spraying equipment. The specific process parameters are as follows:  
1. Rough grinding: grinding speed 60-100m/min, feed speed 150-250mm/rad, grinding depth 1.0-1.5mm.  
2. Thermal spraying: spraying temperature 250-350°C, spraying speed 5-20mm/s, spraying thickness 0.2-0.5mm.  
3. Fine grinding: grinding speed 80-120m/min, feed speed 200-300mm/rad, grinding depth 0.2-0.5mm.  
**Scheme 2:** Rough turning, arc spraying and fine turning. The technologies involved are arc spraying and turning, and the equipment used are lathes and arc spraying equipment. The specific process parameters are as follows:  
1. Rough turning: cutting speed 50-80m/min, feed speed 0.1-0.3mm/rad, depth of cut 0.2-0.5mm.  
2. Arc spraying: spraying temperature 250-350°C, spraying speed 5-20mm/s, spraying thickness 0.2-0.5mm.  
3. Fine turning: cutting speed 80-120m/min, feed speed 0.1-0.3mm/rad, depth of cut 0.1-0.3mm.

**Fig. 5.** Feasible remanufacturing schemes generation.



**Table 6**

The combined assessed value of different schemes.

| Serial number | Scheme                                        | Time   | Cost   | Energy consumption | Combined assessed value |
|---------------|-----------------------------------------------|--------|--------|--------------------|-------------------------|
| 1             | Rough grinding-thermal spraying-fine grinding | 0.5527 | 0.5779 | 0.5272             | 0.5577                  |
| 2             | Rough turning-arc spraying-fine turning       | 0.5966 | 0.5882 | 0.5982             | 0.5936                  |

value and the combined assessed value based on the remanufacturing requirements. The results align closely with actual values, which demonstrates the feasibility and effectiveness of the proposed mechanism.

Using the above methodology, the process designers only need to ask questions based on the failure forms of the recovered products/components, and then set objectives and weights based on the remanufacturing needs to obtain the optimal remanufacturing scheme and process parameters. This approach significantly reduces manual reliance on ODRP and enhances decision-making efficiency.

#### 4.2. Discussion

##### 4.2.1. The effect of the proposed knowledge-augmented strategy

To demonstrate the knowledge reuse effect of the proposed knowledge-augmented strategy, this paper selects the popular models ChatGLM3-6B, Qwen2-7B, and Index-1.9B. Then, based on the characteristics of the automobile gearbox, a set of multiple-choice questions with varying difficulty levels regarding ODRP was compiled. The ChatGLM3-6B that incorporates this knowledge database was compared with other popular LLMs, and the performance was assessed based on the scores of each model on different questions. The question settings and results are shown in [Tables 7 and 8](#), respectively.

From the above results, it can be seen that the knowledge-augmented strategy has a higher accuracy rate compared to the other three models in questions that involve only superficial knowledge of the ODRP for automobile gearboxes. On the other hand, the accuracy of this strategy is

**Table 7**

ODRP question setting.

| Question level             | Question design                                                                                                                                                                    | Option design                                                                                                                            | Number of question |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Simple question (SQ)       | Surface knowledge based on ODRP for automobile gearbox (e. g. direct questions about common terminology, techniques, tools, etc.).                                                 | Provide clear and concise options where one is correct and the others are wrong or irrelevant.                                           | 50                 |
| Medium question (MQ)       | Solve slightly more complex questions based on basic knowledge of ODRP for automobile gearboxes (e.g. standard work procedures or combinations of multiple simple problems, etc.). | The differences between the options are less obvious and require some analysis or reasoning to select the correct answer.                | 30                 |
| Difficulties question (DQ) | Solve complex questions based on in-depth knowledge of the ODRP for automobile gearbox (e.g. detailed parameters of the work steps).                                               | The options provided are confusing and the correct answer needs to be arrived at through in-depth analysis and a combination of factors. | 20                 |

**Table 8**

Accuracy analysis.

| Model                            | Accuracy SQ    | Accuracy MQ    | Accuracy DQ    | ALL            | AVG  |
|----------------------------------|----------------|----------------|----------------|----------------|------|
| ChatGLM3-6B + Knowledge database | 0.96<br>( 48 ) | 0.77<br>( 23 ) | 0.65<br>( 13 ) | 0.84<br>( 84 ) | 0.79 |
| ChatGLM3-6B                      | 0.92<br>( 46 ) | 0.67<br>( 20 ) | 0.35<br>( 7 )  | 0.73<br>( 73 ) | 0.65 |
| Qwen2-7B                         | 0.94<br>( 47 ) | 0.70<br>( 21 ) | 0.40<br>( 8 )  | 0.76<br>( 76 ) | 0.68 |
| Index-1.9B                       | 0.88<br>( 44 ) | 0.60<br>( 18 ) | 0.30<br>( 6 )  | 0.68<br>( 68 ) | 0.59 |

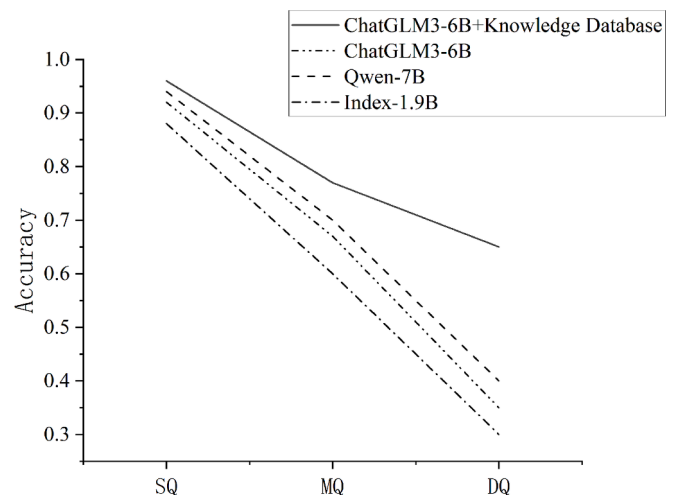
significantly better than the other models in questions that rely on in-depth knowledge or detailed data about the ODRP for automobile gearboxes. It can also be seen that although the accuracy of ChatGLM3-6B, the core generative model selected in this paper, is lower than that of Qwen2-7B in answering the three types of questions, the accuracy of answering the three types of questions is significantly improved after the integration with the remanufacturing process knowledge database, and its average accuracy jumps from 65 % to 79 %. This result indicates that this strategy used effectively improves the performance of the model in answering questions in the specialized domain.

To further illustrate the efficacy of the proposed strategy, the trend of the above model in answering questions of different difficulties was analyzed, as shown in [Fig. 6](#). It can be seen that, with the increase of the difficulty of the question, although the accuracy of each model generally shows a downward trend, the strategy has a flatter downward trend, and the performance is also more outstanding.

The results of the above comparative experiments show that, according to the RAG technology, the performance of relatively weak LLMs can reach or even exceed that of the original strong models by integrating the knowledge database. While the proposed strategy can efficiently reuse knowledge in the remanufacturing process, its knowledge reuse effectiveness may not yet align with that of fine-tuning technology. However, considering the high computational requirements of fine-tuning technology and the risk of catastrophic forgetting, RAG remains a more appropriate choice for remanufacturing enterprises, despite some limitations.

##### 4.2.2. The effect of the proposed function-augmented mechanism

This section will base the discussion on the two remanufacturing process schemes (each consisting of three work steps) proposed in [Section 4.1.1](#), and define the question as “calculating the time/cost/energy consumption required for a specific work step”. This will result in 18



**Fig. 6.** The trend of accuracy rates for different methods in answering questions of varying difficulty.

specific questions. Subsequently, all questions were input into the ChatGLM3-6B to evaluate the accuracy of the function-augmented mechanism in function selection and function calling. Finally, we will obtain the objective values and combined assessed values generated by accurate function selections and calls, and compare these with the actual situation to further evaluate the proposed mechanism's effectiveness in optimal decision-making.

#### (1) The accuracy of function selection and function calling

After organizing the questions, each of which will be input into the ChatGLM3-6B 50 times to evaluate the function-augmented mechanism's accuracy in correctly selecting and calling functions. The results are shown in Figs. 7 and 8.

As shown in Figs. 7 and 8, the accuracy of ChatGLM3-6B in selecting objective functions—such as time, cost, and energy consumption—for six types of work steps ranges from 72 % to 80 %, while the accuracy of calling these functions ranges from 66 % to 72 %.

Additionally, the accuracy of function calling for any work step is typically slightly lower than that of function selection, as function calling follows function selection, and any deviation during function selection can affect the accuracy of subsequent function calling. However, since the proposed mechanism achieves higher accuracy in the function selection phase, the impact of such deviations on function calling is limited. The aforementioned work demonstrates that this function-augmented mechanism accuracy in correctly selecting and calling functions achieves high and stable accuracy, meeting the need to calculate objective values.

#### (2) The accuracy of optimal decision-making

To further illustrate the efficacy of the function-augmented mechanism for ODRP, we will obtain the objective values of schemes 1 and 2

generated by accurate function selections and calls, and compare these with the actual situation. The results are shown in Figs. 9 and 10.

Based on the comparison above, it is evident that the arbitrary objective value generated by the function-augmented mechanism is very close to the actual values. Specifically, the minimum error is 0.0006, and the maximum error is 0.0093. This error range meets the threshold range of practical operational error and provides a sound foundation for the subsequent calculation of the combined assessed values.

Next, the weighted sum of the objective values for each work step is calculated for each scheme to obtain the final combined assessed value of every scheme. And compared the final combined assessed value of each scheme with the actual situation. In accordance with the rationale for the allocation of weights at each level presented in Section 4.1.2, it is similarly assume that the weights  $\lambda_j$  between work steps  $S_{ij}$  in the remanufacturing scheme  $S_i$  are equally distributed, i.e.,  $\lambda_j^1 : \lambda_j^2 : \lambda_j^3 = 1 : 1 : 1$ , and the weights  $\lambda_t$  between remanufacturing objectives in work steps  $S_{ij}$  are  $\lambda_t^1 : \lambda_t^2 : \lambda_t^3 = 4 : 4 : 2$ . The result is shown in Table 9.

Through comparison, it is obvious that the errors between the final combined assessed values of schemes 1 and 2 generated by the function-augmented mechanism and the actual situation are 0.0024 and 0.0029, respectively, which are within the acceptable error range. Additionally, the differences between the combined assessed values of schemes 1 and 2, as generated by the proposed mechanism, and their actual values are 0.0359 and 0.0307, respectively. Clearly, all the selections based on their results correspond to scheme 1.

To sum up, after an in-depth analysis of the actual data and the data generated through the function-augmented mechanism, the results show that both exhibit a high degree of agreement in terms of the consistency of the objective value and combined assessed values. The presence of errors may be attributed to the fact that the adopted function formulas are based on empirical derivation, a phenomenon that is common in engineering practice. Overall, the function-augmented mechanism

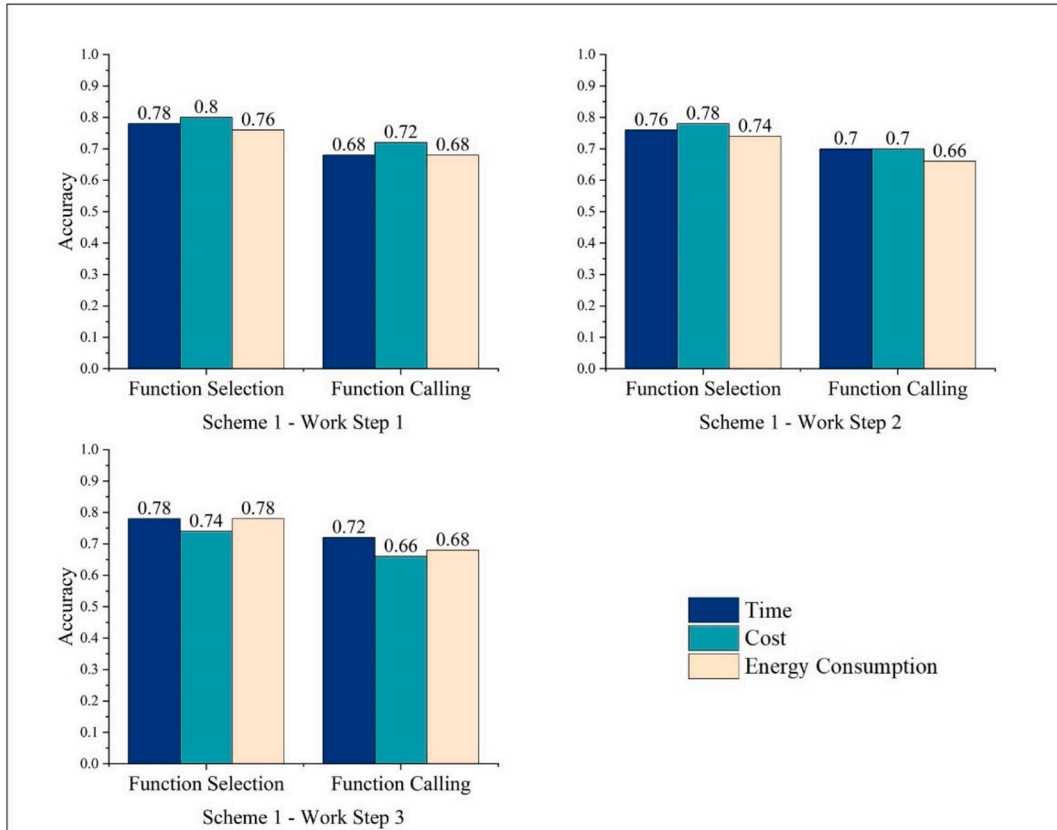


Fig. 7. The accuracy of function selection and function calling (Scheme 1).

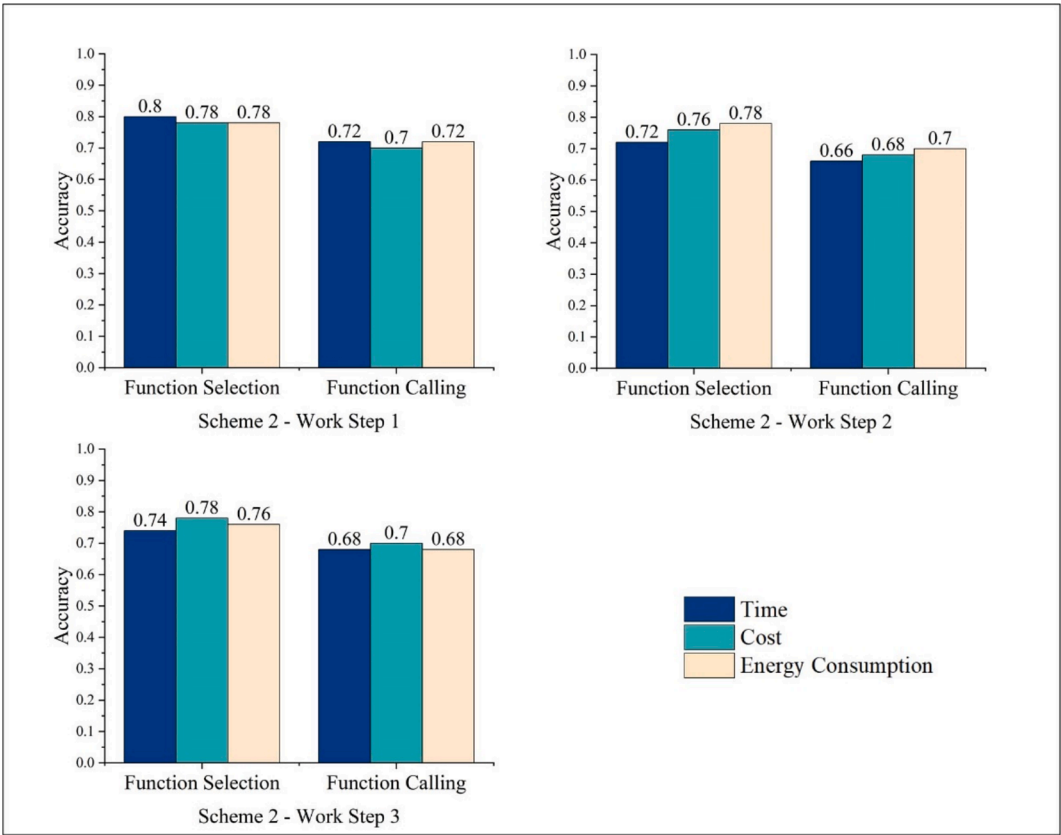


Fig. 8. The accuracy of function selection and function calling (Scheme 2).

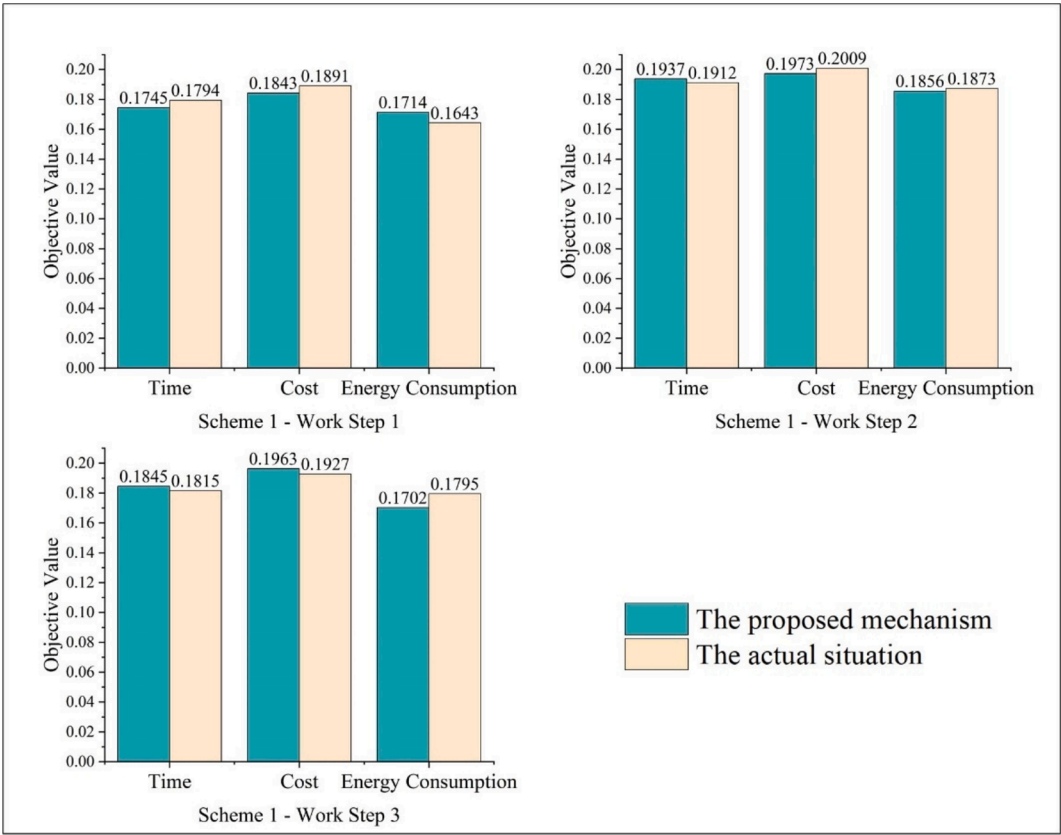


Fig. 9. The objective values of function-augmented mechanism and actual situation (Scheme 1).

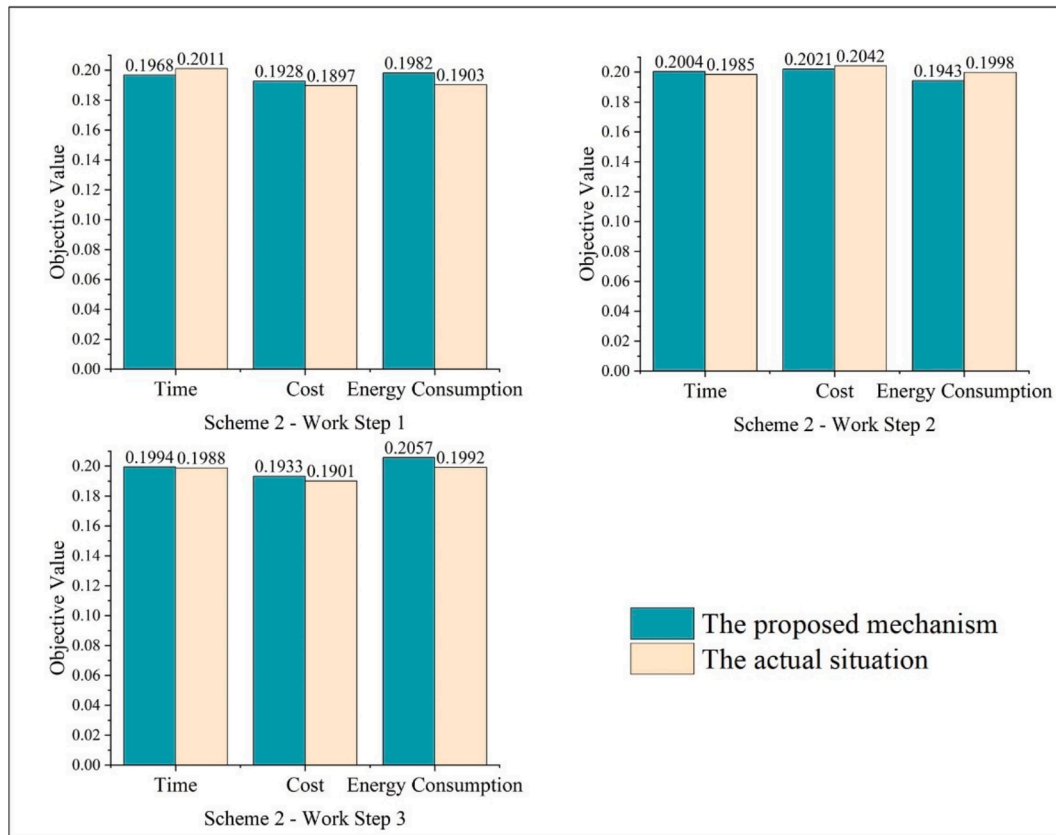


Fig. 10. The objective values of function-augmented mechanism and actual situation (Scheme 2).

Table 9

The combined assessed values of function-augmented mechanism and actual situation.

| Serial number | Form                   | Time   | Cost   | Energy consumption | Weight                                                | Combined assessed value |
|---------------|------------------------|--------|--------|--------------------|-------------------------------------------------------|-------------------------|
| Scheme 1      | The proposed mechanism | 0.5527 | 0.5779 | 0.5272             | $\lambda_j^1 : \lambda_j^2 : \lambda_j^3 = 1 : 1 : 1$ | 0.5577                  |
|               | Actual value           | 0.5521 | 0.5827 | 0.5311             |                                                       | 0.5601                  |
| Scheme 2      | The proposed mechanism | 0.5966 | 0.5882 | 0.5982             | $\lambda_i^1 : \lambda_i^2 : \lambda_i^3 = 4 : 4 : 2$ | 0.5936                  |
|               | Actual value           | 0.5984 | 0.5840 | 0.5893             |                                                       | 0.5908                  |

provides solid data validation for subsequent optimal decision support.

## 5. Conclusions

To overcome the time-consuming and labor-intensive, this paper proposed an LLM-based knowledge and function-augmented approach for ODRP. In this approach, a RAG-based knowledge-augmented strategy is designed to retrieve relevant information and generate feasible remanufacturing schemes. Then, a function-augmented mechanism with function learning is employed to enable LLMs to call the relevant decision objective formulas and calculate the combined assessed values of each feasible remanufacturing scheme, which is helpful to assist process designers to design optimal remanufacturing scheme and process parameters. The proposed approach has the capacity to effectively address the challenges associated with efficient knowledge reuse and multi-objective decision-making challenges of ODRP. Furthermore, it can support the application of LLMs in industry, and offers a novel solution for knowledge-intensive decision-making tasks that has a broad application prospect.

It should be noted that the remanufacturing process knowledge evolves dynamically with the emergence of new materials and technologies, whereas the database in this paper is largely static. In future research, establishing a continuous learning mechanism to address the

challenges of knowledge updates due to new processes, materials, and irregular components is crucial. Furthermore, the failure of EOL components is closely related to their lifecycles, how to obtain their failure form, failure degree and failure timing through multimodal lifecycle data to maximize the benefits of remanufacturing, is also an important issue that needs to be explored in depth in the future.

## CRediT authorship contribution statement

**Haiyang Zhang:** Writing – original draft, Methodology, Conceptualization. **Wei Yan:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Huicong Hu:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Xumei Zhang:** Validation, Data curation. **Qingtao Liu:** Supervision, Writing – review & editing. **Hong Xia:** Investigation. **Yingguang Zhang:** Investigation. **Yuhao Lin:** Investigation.

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## Data availability

Data will be made available on request.

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